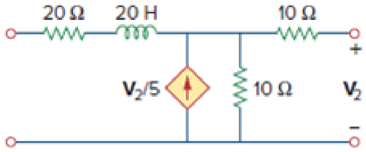
Chapter 19, Problem 7P

Problem

Calculate the *z* parameters of the circuit in Fig. 19.71 as functions of *s*.

Figure 19.71



Step-by-step solution

Step 1 of 10

Refer to Figure 19.71 in the text book.

Connect the voltage source $\mathbf{V}_1(s)$ as shown in Figure 1. Represent the parameters in Laplace notation. Identify the currents and voltages in the circuit.

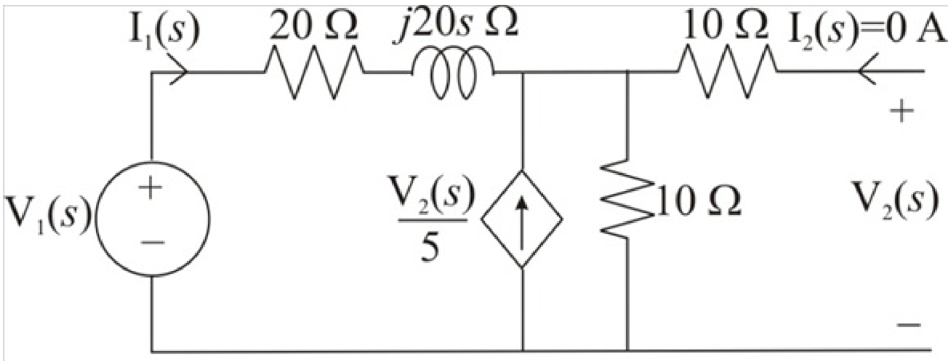


Figure 1

Step 2 of 10

Consider the expression for the voltage $\mathbf{V}_2(s)$.

$$\mathbf{V}_2(s)=\left[\mathbf{I}_1(s)+\frac{\mathbf{V}_2(s)}{5}\right](10)$$
$$\mathbf{V}_2(s)=10\mathbf{I}_1(s)+2\mathbf{V}_2(s)$$
$$2\mathbf{V}_2(s)-\mathbf{V}_2(s)=-10\mathbf{I}_1(s)$$
$$\mathbf{V}_2(s)=-10\mathbf{I}_1(s)$$

Therefore, the expression for the voltage $\mathbf{V}_2(s)$ is $-10\mathbf{I}_1(s)$.

Step 3 of 10

Consider the expression for the voltage $\mathbf{V}_1(s)$.

$$\mathbf{V}_1(s) = \mathbf{I}_1(s)(20+20s) + \left[\mathbf{I}_1(s) + \frac{\mathbf{V}_2(s)}{5}\right](10)$$
$$= \mathbf{I}_1(s)(20+20s) + 10\mathbf{I}_1(s) + 2\mathbf{V}_2(s)$$
$$= \mathbf{I}_1(s)(20+20s+10) + 2\mathbf{V}_2(s)$$

Substitute $-10\mathbf{I}_1(s)$ for $\mathbf{V}_2(s)$.

$$\mathbf{V}_1(s) = \mathbf{I}_1(s)(30+20s) + 2\left[-10\mathbf{I}_1(s)\right]$$
$$= \mathbf{I}_1(s)(30+20s) - 20\mathbf{I}_1(s)$$
$$= \mathbf{I}_1(s)(30+20s-20)$$
$$= \mathbf{I}_1(s)(10+20s)$$

Therefore, the expression for the voltage $\mathbf{V}_1(s)$ is $\mathbf{I}_1(s)(10+20s)$.

Step 4 of 10

Determine the expression for the open circuit input impedance $\mathbf{z}_{11}(s)$.

$$\mathbf{z}_{11}(s) = \frac{\mathbf{V}_1(s)}{\mathbf{I}_1(s)}$$

Substitute $\mathbf{I}_1(s)(10+20s)$ for $\mathbf{V}_1(s)$.

$$\mathbf{z}_{11}(s) = \frac{\mathbf{I}_1(s)(10+20s)}{\mathbf{I}_1(s)}$$
$$= 10+20s\ \Omega$$

Therefore, the expression for the open circuit input impedance $\mathbf{z}_{11}(s)$ is $\boxed{10+20s\ \Omega}$.

Step 5 of 10

Determine the value of the open circuit transfer impedance $\mathbf{z}_{21}(s)$ from port-2 to port-1.

$$\mathbf{z}_{21}(s) = \frac{\mathbf{V}_2(s)}{\mathbf{I}_1(s)}$$

Substitute $-10\mathbf{I}_1(s)$ for $\mathbf{V}_2(s)$.

$$\mathbf{z}_{21}(s) = \frac{-10\mathbf{I}_1(s)}{\mathbf{I}_1(s)}$$
$$= -10\ \Omega$$

Therefore, the value of the open circuit transfer impedance $\mathbf{z}_{21}(s)$ from port-2 to port-1 is $\boxed{-10\ \Omega}$.

Step 6 of 10

Connect the voltage source $\mathbf{V}_2(s)$ as shown in Figure 2. Identify the currents and voltages in the circuit.

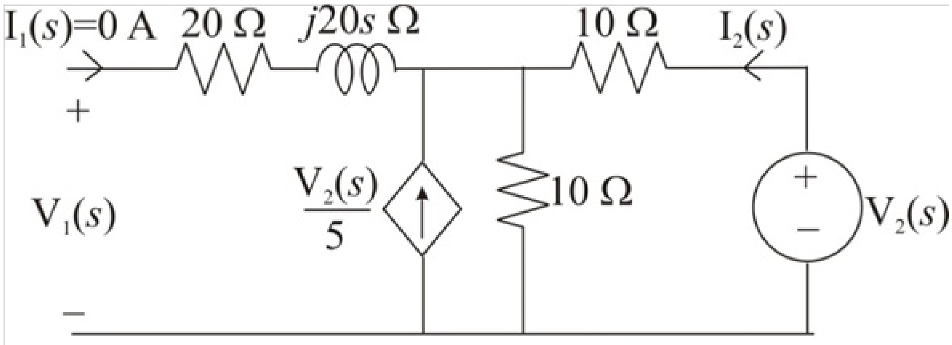


Figure 2

Step 7 of 10

Consider the expression for the voltage $\mathbf{V}_2(s)$.

$$\mathbf{V}_2(s) = \mathbf{I}_2(s)(10) + \left[\mathbf{I}_2(s) + \frac{\mathbf{V}_2(s)}{5}\right](10)$$
$$\mathbf{V}_2(s) = \mathbf{I}_2(s)(10) + 10\mathbf{I}_2(s) + 2\mathbf{V}_2(s)$$
$$2\mathbf{V}_2(s) - \mathbf{V}_2(s) = -20\mathbf{I}_2(s)$$
$$\mathbf{V}_2(s) = -20\mathbf{I}_2(s)$$

Therefore, the expression for the voltage $\mathbf{V}_2(s)$ is $-20\mathbf{I}_2(s)$.

Step 8 of 10

Consider the expression for the voltage $\mathbf{V}_1(s)$.

$$\begin{aligned}\mathbf{V}_1(s) &= \left[\mathbf{I}_2(s) + \frac{\mathbf{V}_2(s)}{5}\right](10)\\ &= 10\mathbf{I}_2(s) + 2\mathbf{V}_2(s)\end{aligned}$$

Substitute $-20\mathbf{I}_2(s)$ for $\mathbf{V}_2(s)$.

$$\begin{aligned}\mathbf{V}_1(s) &= 10\mathbf{I}_2(s) + 2\left[-20\mathbf{I}_2(s)\right]\\ &= 10\mathbf{I}_2(s) - 40\mathbf{I}_2(s)\\ &= -30\mathbf{I}_2(s)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_1(s)$ is $-30\mathbf{I}_2(s)$.

Step 9 of 10

Determine the value of the open circuit transfer impedance $\mathbf{z}_{12}(s)$ from port-1 to port-2.

$$\mathbf{z}_{12}(s)=\frac{\mathbf{V}_1(s)}{\mathbf{I}_2(s)}$$

Substitute $-30\mathbf{I}_2(s)$ for $\mathbf{V}_1(s)$.

$$\begin{aligned}\mathbf{z}_{12}(s)&=\frac{-30\mathbf{I}_2(s)}{\mathbf{I}_2(s)}\\&=-30\ \Omega\end{aligned}$$

Therefore, the value of the open circuit transfer impedance $\mathbf{z}_{12}(s)$ from port-1 to port-2 is $\boxed{-30\ \Omega}$.

Step 10 of 10

Determine the expression for the open circuit output impedance $\mathbf{z}_{22}(s)$.

$$\mathbf{z}_{22}(s)=\frac{\mathbf{V}_2(s)}{\mathbf{I}_2(s)}$$

Substitute $-20\mathbf{I}_2(s)$ for $\mathbf{V}_2(s)$.

$$\begin{aligned}\mathbf{z}_{22}(s)&=\frac{-20\mathbf{I}_2(s)}{\mathbf{I}_2(s)}\\&=-20\ \Omega\end{aligned}$$

Therefore, the expression for the open circuit output impedance $\mathbf{z}_{22}(s)$ is $\boxed{-20\ \Omega}$.